Elementary Problems
Operation, Game and Invariance
Solutions

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Basic Work

1. We prove this by induction on n. For $n = 1$, since we can change the state of the only light bulb with label 1, we can always make it 'off'.

   For the inductive step, we first change the states of the last two light bulbs if the light bulb with label n is 'on'. So we may assume that light bulb is 'off'. By the inductive hypothesis, we can change the states of the first $n - 1$ light bulbs so that they are all 'off'. So we are done by induction.

2. Yes, the process must come to an end. We prove this by induction on the number of cards. We use W (and B) to denote a card with a white (black, respectively) side facing upwards. When $n = 2$, the only initial configuration that we may proceed is WW. After one step, it becomes BB and the process ends.

   Assume the claim holds when there are $k \geqslant 2$ cards. Consider the case when there are $k+1$ cards. If we never flip the $(k+1)$st card, then the case is identical to the case when there are k cards. Thus, we are done. Therefore, we may assume the $(k+1)$st card is flipped in a certain step. This can only happen when the $(k+1)$st card is changed from white to black, since otherwise we cannot choose this card. Afterwards, the $(k+1)$st card can no longer be chosen. The case is again identical to the case when there are k cards. By the inductive hypothesis, the process must come to an end.

   Hence, the process must end in finitely many steps by induction.

3. From the construction, it is obvious that any two consecutive words differ in exactly one digit. We prove that the sequence generates each word once by induction on n. When $n = 1$, the sequence is $0, 1$. Assume this holds for some case $n = k$ with $k \geqslant 1$. Consider the case $n = k + 1$.

   Since $e(1), e(2), \ldots, e(2^k - 1)$ are less than k, the last digits (meaning the $(k+1)$st digits) of the words $w_0, w_1, \ldots, w_{2^k-1}$ are not changed. Also, the changes applied to the first k digits are identical to the case $n = k$. Thus, by the inductive hypothesis, these 2^k words cover all possibilities of words of length $k + 1$ with the last digits 0.

   Next, as $e(2^k) = k$, w_{2^k} is obtained by changing the last digit of w_{2^k-1} from 0 to 1.

   For $1 \leqslant j \leqslant 2^k - 1$, we have $e(2^k + j) = e(j)$. Therefore, the subsequent operations do not alter the last digit. Also, the changes are again identical to the case $n = k$. This shows the words $w_{2^k}, w_{2^k+1}, \ldots, w_{2^{k+1}-1}$ cover all possibilities of words of length $k + 1$ with the last digits 1. (Strictly speaking, the inductive hypothesis only applies when the initial word is $00\cdots0$. However, if we can obtain any word starting from $00\cdots0$, then clearly we can obtain any word starting from an arbitrary word as the operation as nothing to do with the exact values of the digits.)

   Now, we find that every word of length $k + 1$ has been generated. The assertion then follows by induction.

4. Note that we can assume at most one operation is used in each row or column. We prove the assertion by induction. Consider the base case $n = 4$. Suppose it is possible to make at most 3 copies of $+$. WLOG assume all signs in the first row are $-$. This can only be done by either changing the first column but not the other columns or the first row, or changing the first row and only the columns except the first. In both cases as shown below, no matter the remaining three rows are operated or not, there are exactly 6 copies of $+$. This is a contradiction. So the base case holds.

$-$	$-$	$-$	$-$
$+$	$+$	$-$	$-$
$+$	$-$	$+$	$-$
$+$	$-$	$-$	$+$

$-$	$-$	$-$	$-$
$-$	$-$	$+$	$+$
$-$	$+$	$-$	$+$
$-$	$+$	$+$	$-$

Assume the case $n-1$ holds. We consider the inductive step with an $n \times n$ chart. Then the bottom right $(n-1) \times (n-1)$ sub-chart always contain at least $n-1$ copies of $+$ by the inductive hypothesis. The assertion is true if the first row or the first column contains a $+$. Therefore, it suffices to consider the case when all signs in the first row and the first column are $-$. Up to symmetry, this can only be done by changing only the first column and all the rows except the first. But then there are exactly $(n-2)(n-1) > n$ copies of $+$. So the inductive step is established.

By induction, there are always n or more $+$ signs.

5. The answer is 7.

We first show that the number of questions asked cannot be less than 7. By asking a question, there are 2 possible outcomes. One of the outcomes can only allow Y to reject at most half of the remaining possibilities of the value of n. This means the number of possibilities is at least half of that before the question is asked. Therefore, if Y only askes 6 questions, then it is possible that the number of possibilities is at least $\left\lceil \frac{100}{2^6} \right\rceil = 2$. In that case, Y cannot deduce the value of n.

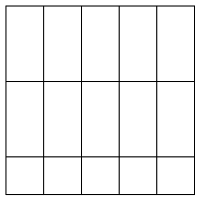
Next, we show that 7 questions suffice. We shall show that Y can determine $n \pmod{2^k}$ by asking k questions. For the base case $k = 1$, Y can simply take $a = b = 2$. If the answer is 'yes', then n is even. If the answer is 'no', then n is odd.

Assume Y finds that $n \equiv t \pmod{2^k}$ for some $k \geqslant 1$ after asking k questions. WLOG assume $0 \leqslant t < 2^k$. Next, Y can take $a = 2^k - t$ and $b = 2^{k+1}$. If the answer is 'yes', then $n \equiv t + 2^k \pmod{2^{k+1}}$. If the answer is 'no', then $n \equiv t \pmod{2^k}$. Inductively, Y can determine $n \pmod{2^k}$ by asking k questions. In particular, Y can determine $n \pmod{128}$ by asking 7 questions. Since n lies between 1 and 100, this uniquely determines the value of n.

Therefore, the minimum number of questions needed is 7.

6. The answer is 6. Let S be the score of the first player.

   We first show that the second player can ensure $S \leqslant 6$. We partition the 5×5 board as shown below. Whenever the first player enters a 1 into a square belonging to a 2×1 domino, the second player should enter a 0 into the square belonging to the same domino if it is empty. Other than that, he can enter a 0 randomly. This ensures every domino contains at least one 0. As every 3×3 square contains three dominoes completely, there are at least three 0's inside. Thus, $S \leqslant 6$.

   Secondly, we show that the first player can ensure $S \geqslant 6$. We label the rows and columns in the usual order and use (i, j) to denote the square in the ith row and jth column. Initially, the first player enters a 1 into $(3, 3)$. WLOG we may assume the second player enters a 0 in the 4th or the 5th column. In the subsequent rounds, the first player can enter the number into $(2, 1), (2, 2), (2, 3), (3, 1), (3, 2)$ until all these squares are occupied. Note that he can enter at least three numbers into these squares. Therefore, after four rounds, there are four 1's in the squares $(2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)$. Meanwhile, the second player has entered at most three 0's in the first three columns. We claim that one of the 3×3 squares

   $$A = \{(i, j) : 1 \leqslant i \leqslant 3,\ 1 \leqslant j \leqslant 3\}$$

   and

   $$B = \{(i, j) : 2 \leqslant i \leqslant 4,\ 1 \leqslant j \leqslant 3\}$$

   has at least 3 empty squares. If not, then each of them has at most 2 empty squares. Then at least $12 - 2 - 2 = 8$ squares in the first three columns are occupied. This contradicts the above observation as there can be at most 7 occupied squares in the first three columns. WLOG assume A has at least 3 empty squares. Then the first player can enter at least 2 more 1's in A so that A contains at least six 1's. Thus, $S \geqslant 6$.

   From the strategies of the two players, we conclude that the largest S is 6.

7. Suppose on the contrary that Vasya has a winning strategy (by taking the last stone). For each $k = 1, 2, \ldots, n + 1$, Peter loses by taking away kn stones in the first step. The strategy of Vasya's won't be taking away $k'n$ stones, since otherwise Peter can win by taking away $(k + k')n$ stones at the beginning. Suppose Vasya can win by taking away

p_k stones where p_k is a prime less than n or $p_k = 1$. As there are at most n possibilities for p_k but $n+1$ choices for k, there exist distinct k and ℓ such that $p_k = p_\ell = p$.

WLOG assume $k < \ell$. This shows after removing $kn + p$ stones, the second player has a winning strategy. Similarly, if exactly $\ell n + p$ stones are removed, Vasya still has a winning strategy. But in that case, Peter could indeed take away $(\ell - k)n$ stones, leaving exactly $kn + p$ stones. This is the situation above for which the second player, who is now Peter, has a winning strategy. This is a contradiction.

Thus, Peter can take the last stone.

Algorithm

8. Yes. For the first command, 1 and 2 swap their places, while k and $n+3-k$ swap their places for $k = 3, 4, \ldots, \left\lfloor \dfrac{n+2}{2} \right\rfloor$. After the first command, the positions become $2, 1, n, n-1, n-2, \ldots, 3$.
   For the second command, 2 and n swap their places, while k and $n+2-k$ swap their places for $k = 3, 4, \ldots, \left\lfloor \dfrac{n+1}{2} \right\rfloor$. Then the positions become $n, 1, 2, 3, \ldots, n-1$ as desired.

9. X has a winning strategy. Indeed, X can choose 99 in the first step. Afterwards, we partition the remaining integers as follows:
   $$\{1, 2\}, \{3, 4\}, \ldots, \{97, 98\}.$$
   The integers in the same pair are relatively prime. Therefore, whenever Y chooses an integer k, X can choose the other integer in the same pair as k so that they are relatively prime. This shows X can always choose a suitable integer. As the game must come to an end, Y must lose eventually.

10. Y has a winning strategy. Consider the following pairing of the numbers:
    $$\{1, 30\}, \{2, 29\}, \ldots, \{15, 16\}.$$
    The sum of integers in each pair is 31, which is a prime number. Therefore, whenever X chooses an integer k, Y can choose the other integer in the same pair as k so that the sum is 31. This shows Y can always choose a suitable integer. As the game must come to an end, X must lose eventually.

11. Y has a winning strategy. The strategy is to apply the rotational symmetry. In round 1, whenever X marks a point A, Y can mark a point A' so that AA' is a diameter of the circle. Such a point has not been marked since all marked points can be paired up to form some diameters.

In round 2, whenever X joins the chord AB, Y can join the chord $A'B'$ so that AA' and BB' are diameters of the circle. From the above construction, if A, B are red, then A', B' must be blue. Also, such a chord has not been joined. If $A'B'$ intersects a red chord CD in the interior of the circle, then $C'D'$ (where CC' and DD' are diameters of the circle) is blue and it has been joined before the chord AB is joined. Also, $C'D'$ intersects AB in the interior. This shows the operation of X violates the rule, and hence this cannot happen. Therefore, Y can always perform the operation whenever X can. As there is a finite number of chords, the game must end and the one who loses must be X.

12. X has a winning strategy. Firstly, X chooses an even number. Afterwards, X always chooses a number which has the same parity as the one chosen by Y until this cannot be done. Since there are 617 odd numbers and 616 even numbers except the first one chosen by X, if X can always use this strategy, then X must choose 308 odd numbers and 309 even numbers at the end. The sum is even.

    Suppose X cannot use this strategy after some rounds. Then all odd numbers have been chosen. Thus, X has chosen 308 odd numbers. All other numbers chosen afterwards must be even. So the sum remains to be even, and X has a winning strategy.

13. (a) We label the boxes from 1 to m. Consider the following sequence of operations:

$$\begin{aligned}
&\text{putting 1 ball in each of } 2, 3, \ldots, n+1,\\
&\text{putting 1 ball in each of } 2, 3, \ldots, n+1,\\
&\text{putting 1 ball in each of } 3, 4, \ldots, n+2,\\
&\text{putting 1 ball in each of } 4, 5, \ldots, n+3,\\
&\ldots\\
&\text{putting 1 ball in each of } m, 1, 2, \ldots, n-1.
\end{aligned}$$

    Then the number of balls in the box 1 is increased by $n-1$, that in the box $n+1$ is increased by $n+1$, while those in the boxes $2, 3, 4, \ldots, n, n+2, n+3, \ldots, m$ are increased by n. Since we are only concerned about the differences of the number of balls in the boxes, this has the same effect as taking 1 ball from box 1 and put it in box $n+1$. Due to symmetry, we can transfer 1 ball from any box to another box.

    Since $(m, n) = 1$, the linear polynomial kn runs through all residue classes modulo m where $k \in \mathbb{N}$. Thus, we can carry out a finite number of operations to make the total number of balls a multiple of m. Thus, we may assume the total number of balls is md for some $d \in \mathbb{N}$.

    We can use the above sequence of operations to transfer a ball from a box with more than d balls to a box with fewer than d balls. By repeating the same procedure, we can make each box contain the same number of balls. This completes the proof.

14. The first player X has a winning strategy. Divide the board into 25 copies of 3×4 sub-boards. X first puts a 1×2 domino in the centre of one sub-board. Each time after

the second player Y places a 2×1 domino, this domino either lies in a sub-board which contains a 1×2 domino or not. In the former case, X simply puts a 1×2 domino in the centre of an empty sub-board if this is possible. In the latter case, this 1×2 domino must be the only domino being put in this sub-board (by X or Y) under our strategy. Also, it only occupies one half, either left or right, of the sub-board. So X can put a 1×2 domino in the middle row of another half of the same sub-board.

The same strategy applies until there is no empty sub-board. Under this strategy, it is clear that each sub-board must contain exactly one domino put by X, and each such domino is put in the middle row of the board. For each of the 25 dominoes put by X, there are 2 other vacancies of 1×2 squares available in the same columns in which X can put more dominoes as it is clear that those squares cannot be occupied by any 2×1 dominoes. Therefore, X is guaranteed to put at least $25 \times 3 = 75$ dominoes in total. However, each 2×1 domino put by Y must have one cell lying in the middle row. As 50 cells in the middle row have been occupied by X's dominoes, Y can put at most 50 dominoes in total. Hence, X wins under this strategy.

15. Yes. We shall prove by induction on n that B can always meet the goal. The case $n = 1$ is trivial. Assume this is true for some case $n = k \geqslant 1$. Consider the case when there are $k + 1$ cells.

Applying the inductive hypothesis, B can make the first k cells be in monotonic increasing order. Suppose the numbers are

$$a_1 \leqslant a_2 \leqslant \cdots \leqslant a_k \quad \text{and} \quad a_{k+1}$$

in order. Let x be the number announced by A. If $x \geqslant a_k$, B can replace a_{k+1} by x so that the sequence is monotonic increasing.

If $x < a_k$, then there must be an index $0 \leqslant j \leqslant k$ such that $a_j \leqslant x < a_{j+1}$, where we have defined $a_0 = 0$ for convenience. In that case, B can replace a_{j+1} by x. Afterwards, the first n numbers are still in increasing order as

$$a_1 \leqslant a_2 \leqslant \cdots \leqslant a_j \leqslant x < a_{j+2} \leqslant \cdots \leqslant a_k.$$

Note that the sum of the first k numbers is decreased as $x < a_{j+1}$. Afterwards, B can repeat the same strategy again and again. Since the sum of the first k numbers is at least 0, it cannot decrease indefinitely. Thus, A must need to choose some x with $x \geqslant a_k$ in some step. Then B can meet the goal. The claim follows by induction.

16. The answer is 4005.

Firstly, we show that 4005 questions are sufficient. Label the boxes from 1 to 2004. We first choose the following pairs of boxes in the first 2003 questions respectively:

$$(1, 2), (1, 3), \ldots, (1, 2004).$$

If one of the answers is 'no', then box 1 does not contain a white ball. This means if the pair $(1, k)$ gives an answer 'yes', then box k contains a white ball. Since there are at least two white balls, we must be able to specify two boxes with white balls. If all answers to the 2003 questions are 'yes', then box 1 must contain a white ball, since otherwise all the other 2003 boxes have to contain a white ball each, yielding an odd number of white balls. Then we choose the following pairs of boxes in the next 2002 questions respectively:

$$(2,3),(2,4),\ldots,(2,2004).$$

Again, if one of the answers is 'no', then box 2 does not contain a white ball, and we can specify all boxes with white balls. If the answers are all 'yes', then box 2 must contain a white ball, since otherwise all the other 2002 boxes have to contain a white ball each, yielding an odd number of white balls (including the one in box 1). Now, both boxes 1 and 2 have white balls and we are done in all cases.

Secondly, we prove that 4004 questions are not sufficient. Suppose we always get the answer 'yes' no matter which boxes we choose, and we are able to specify two boxes, say boxes 1 and 2, containing white balls. If we have never chosen the pair of boxes $(1, 2)$ in the 4004 questions, then it could happen that boxes 1 and 2 do not contain white balls and all other boxes do. This is a contradiction as this gives a case for which boxes 1 and 2 do not contain white balls.

If we have chosen the pair of boxes $(1, 2)$ in some question, then one of the pairs of boxes

$$(1,3),(1,4),\ldots,(1,2004),(2,3),(2,4),\ldots,(2,2004)$$

has not been chosen since there are 4004 pairs but only 4003 questions are left. WLOG assume $(1, 3)$ has not been chosen. Then it could happen that boxes 1 and 3 do not contain white balls and all other boxes do. Again, this is a contradiction.

Therefore, the minimum number of questions needed is 4005.

17. We prove this by induction on n. When $n = 1$, Bob can simply ask if t is on the table. Only 1 move is needed.

    Suppose at most $3n - 1$ moves are needed for the case n. Consider the case when there are $n + 1$ distinct integers with sum S. Firstly, Bob can ask whether $S - 2t$ is on the table.

    - If $S - 2t$ is on the table, then Bob can erase this number in the second move. The remaining n distinct integers have sum $2t$. In the third move, Bob asks if the numbers can be partitioned into two sets with equal sum. If yes, then the numbers in one set has sum t as desired. If not, then any numbers with sum t must include the integer $S-2t$. So it suffices to check whethere there exist some numbers among the remaining n integers with sum $t - (S - 2t) = 3t - S$. This can be done in $3n - 1$ moves by the inductive hypothesis. So $3n + 2$ moves suffice.

- If $S - 2t$ is not on the table, then Bob can add the number $S - 2t$ in the second move. This gives $n + 2$ integers with sum $2S - 2t$. In the third move, Bob asks if the numbers can be partitioned into two sets with equal sum. If yes, then the numbers in the set with $S - 2t$ has sum $S - t$. By removing $S - 2t$, we know that the remaining numbers have sum t. If not, then there cannot be numbers with sum t since otherwise we can simply put those numbers with $S - 2t$ to form a set. So 3 moves suffice.

By induction, Bob can meet the goal in at most $3n - 1$ moves when there are n integers initially.

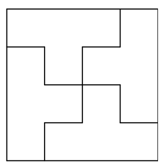
Tiling and Colouring

18. We colour the cells black and white alternately. Note that the two cells removed are of the same colour, say black. Then there are 32 white cells and 30 black cells remaining. As each domino covers exactly 1 black cell and 1 white cell, it is not possible to use 31 dominoes to tile the remaining cells.

19. The answer is $n = 4k$ for some positive integer k.

    Since each T-tetromino consists of 4 cells, n^2 has to be a multiple of 4. This means n is even.

    If $n = 4k$, then we can tile the table with k^2 copies of 4×4 squares formed by T-tetrominoes as shown in the obvious way.



    If $n = 4k+2$, we colour the cells black and white alternately. Note that each T-tetromino covers an odd number of black cells and an odd number of white cells. Since there are

$$\frac{n^2}{4} = (2k + 1)^2$$

    T-tetrominoes in total, they cover an odd number of black cells together. This is a contradiction since there are $\dfrac{n^2}{2} = 2(2k + 1)^2$ black cells.

    Hence, such a tiling exists if and only if n is a multiple of 4.

20. We colour the cells black and white alternately. Since there are $4 \times 5 = 20$ cells in total, there must be 10 black cells and 10 white cells. However, a T-tetromino covers an odd number of black cells but all other tetrominoes cover an even number of black cells. This shows it is impossible for them to cover exactly 10 black cells. So it is impossible to form a rectangle.

21. We colour the squares as shown using colours $1, 2, 3$. For $j = 1, 2, 3$, let a_j be the number of squares of colour j whose lightning bugs are on. Initially, one of a_1, a_2, a_3 is 1 and the others are 0. WLOG assume $a_1 = 1$.

    In each move, for each colour, exactly one lightning bug in a square of that colour changes state. Therefore, each of a_1, a_2, a_3 is increased by 1 or is decreased by 1. Thus, a_1 and a_2 must have different parities. So it is not possible to make $a_1 = a_2 = a_3 = 0$ at the end.

1	2	3	1	2	3
2	3	1	2	3	1
3	1	2	3	1	2
1	2	3	1	2	3
2	3	1	2	3	1
3	1	2	3	1	2

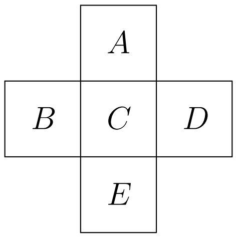
22. (a) Yes. Indeed, we just need to consider modulo 4 of the numbers. There are 16 numbers in each of the residue classes. The following gives one possible way to assign the numbers. Due to symmetry, we just need to check

$$0 + 1 + 1 + 2 \equiv 0 + 2 + 3 + 3 \equiv 1 + 2 + 2 + 3 \equiv 0 + 0 + 1 + 3 \equiv 0 \pmod 4.$$

0	3	2	1	0	3	2	1
1	2	3	0	1	2	3	0
2	1	0	3	2	1	0	3
3	0	1	2	3	0	1	2
0	3	2	1	0	3	2	1
1	2	3	0	1	2	3	0
2	1	0	3	2	1	0	3
3	0	1	2	3	0	1	2

    (b) No. Consider any 5 cells forming a horizontal cross as shown. Both $A+B+C+D$ and $A+B+C+E$ are multiples of 4. So $D \equiv E \pmod 4$. Similarly, we obtain $A \equiv B \equiv D \equiv E \pmod 4$ by considering different orientations of the shape.

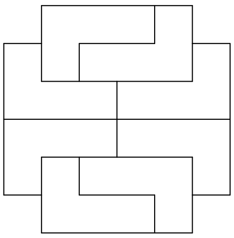
    We colour the cells black and white alternately. From above, any two cells (except the corner cells) which are diagonally adjacent to each other are congruent modulo 4. It follows that any two cells of the same colour are congruent modulo 4, possibly except the 4 corner cells. Thus, the chessboard consists of $\dfrac{8^2-4}{2} = 30$ numbers belonging to the same residue class modulo 4. This is impossible.

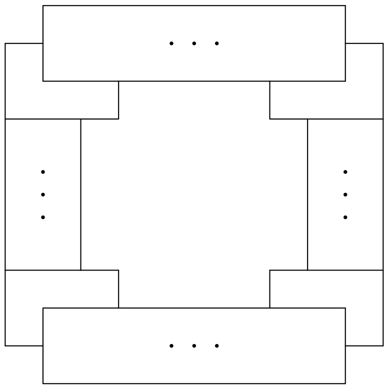


23. The answer is $n = 4k+2$ for some positive integer k.

    Firstly, the total number of cells $n^2 - 4$ has to be a multiple of 4. So n is even. We colour the cells of the table such that the cells in the odd rows are black and the cells in the even rows are white. Note that each L-tetromino covers exactly 1 or 3 black cells. As the total number of black cells and the total number of white cells are the same, the number of tetrominoes covering 1 black cells must be equal to the number of tetrominoes covering 3 black cells. In particular, there is an even number of L-tetrominoes. It follows that $n^2 - 4$ is a multiple of 8, and hence n is of the form $4k+2$.

    We now prove by induction that there exists a tiling when $n = 4k+2$. The base case $n = 6$ is shown below on the left.





We now consider the inductive step. We tile the board of size $n = 4k+2$ as shown above on the right. The middle shape of size $4k-2$ can be tiled using the inductive hypothesis. For the 2 rectangles of size $2 \times 4k$ and the 2 rectangles of size $4(k-1) \times 2$, we can use some 2×4 rectangles formed by two L-tetrominoes to tile these rectangles. So we can tile such a table by induction when n is of the form $4k+2$.

Invariance

24. If $n \equiv 0 \pmod{k+1}$, then Y has a winning strategy. Indeed, whenever X takes away m coins, Y can take away $k+1-m$ coins (note that if $1 \leqslant m \leqslant k$, then $1 \leqslant k+1-m \leqslant k$). The total number of coins taken away by X and Y is $k+1$. Therefore, the number of coins after Y's turn is always a multiple of $k+1$, while that after X's turn is not. As 0 is a multiple of $k+1$, it must be Y who takes away the last coin since the game must come to an end.

    If $n \equiv r \not\equiv 0 \pmod{k+1}$, then X has a winning strategy. WLOG assume $1 \leqslant r \leqslant k$. Then X can take away r coins in the first step. Afterwards, the number of coins is a multiple of $k+1$, and X can follow the same strategy as described above to guarantee a win.

25. If $3 \mid n$, then Y has a winning strategy. Note that $3 \nmid 2^t$ for any t. Therefore, the number of coins N after X's turn is not divisible by 3. If $N \equiv 1 \pmod 3$, then Y can take away 1 coin. If $N \equiv 2 \pmod 3$, then Y can take away 2 coins. In any case, the number of coins after Y's turn must be divisible by 3. As 0 is a multiple of 3, it must be Y who takes away the last coin since the game must come to an end.

    If $3 \nmid n$, then X has a winning strategy. Indeed, if $n \equiv 1 \pmod 3$, then X can take away 1 coin. If $n \equiv 2 \pmod 3$, then X can take away 2 coins. Afterwards, the number of coins is a multiple of 3, and X can follow the same strategy as described above to guarantee a win.

26. We claim that X has a winning strategy if and only if $n \equiv 1, 3, 4, 5, 6 \pmod 7$. This can be proved via induction. The base case $n = 0$ is trivial.

    Assume the claim holds for all smaller cases. Consider the case n. Let N be the total number of coins at a certain stage.

    If $n \equiv 1 \pmod 7$, then X can take away 1 coin in the first step. Then $N \equiv 0 \pmod 7$. From the inductive hypothesis, the second player, who is now player X, has a winning strategy.

    Similarly, if $n \equiv 3, 4, 5, 6 \pmod 7$, then X can take away $1, 4, 3, 4$ coins respectively in the first step. Then $N \equiv 2, 0, 2, 2 \pmod 7$ respectively. From the inductive hypothesis, the second player, who is player X, has a winning strategy.

    If $n \equiv 2 \pmod 7$ and X takes away $1, 3, 4$ coins respectively in the first step, then $N \equiv 1, 6, 5 \pmod 7$ respectively. From the inductive hypothesis, the first player, who is player Y, has a winning strategy in each case.

    If $n \equiv 0 \pmod 7$ and X takes away $1, 3, 4$ coins respectively in the first step, then $N \equiv 6, 4, 3 \pmod 7$ respectively. From the inductive hypothesis, the first player, who is player Y, has a winning strategy in each case.

    Therefore, the claim is true by induction.

27. B has a winning strategy. Indeed, B only needs to take away 1 stone every time. When it is A's turn, the number of stones is odd. Thus, A has to remove an odd number of stones, leaving an even number of stones. After B has taken away 1 stone, the number of stones is again odd. In particular, there is at least 1 stone left after B's turn. Therefore, B will never lose under this strategy. Since the number of stones is strictly decreasing, the game must come to an end, and it must be A who takes away the last stone and loses.

28. Note that $|a-b| \equiv a-b \equiv a+b \pmod 2$. Therefore, the parity of the sum of integers on the blackboard remains unchanged. As

$$1+2+\cdots+(4n+2) = \frac{(1+(4n+2))(4n+2)}{2} = (4n+3)(2n+1)$$

is odd, the integer left must be odd, and hence it cannot be 0.

29. The answer is $\dfrac{n^2+n+2}{2}$.

Each time when we change $+k$ to $-k$, the sum is decreased by $2k$. Therefore, the sum always has the same parity as $S = 1+2+\cdots+n$. The answer is maximized when we choose positive signs for all numbers. In that case it is equal to S. The answer is minimized when we choose negative signs for all numbers, which gives $-S$ as the result. Therefore, only the $\dfrac{S-(-S)}{2}+1 = S+1$ integers between $-S$ and S which have the same parity as S are possible answers. We claim that all of these may happen.

Initially, the sum is S. We carry out the following operation. Each time if the sign of 1 is positive, then we change it to negative. The sum is decreased by 2. If the sign of 1 is negative and the signs are not all negative, then we can always find a number k such that the sign of k is negative while that of $k+1$ is positive. Then we change the signs of k and $k+1$. The sum is decreased by $2(k+1)-2k = 2$. We can carry out the same process repeatedly, until all the signs are negative. Since the sum is decreased by 2 in each step, the result covers all possibilities among $S, S-2, S-4, \ldots, -S$.

Therefore, we can form $S+1 = \dfrac{n(n+1)}{2}+1 = \dfrac{n^2+n+2}{2}$ different answers.

30. The total number of coins a in the piles with odd labels is $n+2$, while the total number of coins b in the piles with even labels is n. In each step, both a and b are increased by 1. Thus, we always have $a \neq b$. However, if all piles have the same number of coins, then we should have $a = b$. So this can never happen.

31. For each 2×2 square, we count the number a_j of black cells. Let S be the sum of all a_j's taken over all 2×2 squares of the chessboard. For each cell not on the side of the chessboard, it belongs to 4 copies of 2×2 squares. So it contributes an even number to S. For each cell on the side but not the corner, it belongs to 2 copies of 2×2 squares. Again it contributes an even number to S. For each corner cell, it belongs to 1 copy of

2×2 squares. The 3 black corner cells each contributes 1, while the white corner cell contributes nothing. Therefore, S must be odd. So there exists a 2×2 square which contributes an odd number to S, which means it consists of an odd number of black cells.

32. The goal can be met if k is odd, or if both n and k are even.

    Firstly, consider the exceptional case where n is odd and k is even. We count the number N of cards having their white sides facing upwards. Initially, $N = n$ is odd. In each step, as an even number of cards are flipped, and each flip changes N by 1, the parity of N remains unchanged. Therefore, it is not possible to make N equal to 0. Thus, there is always a white card.

    For odd k, we proceed by induction on n. The base case $n = k$ is trivial. Suppose the case n is done. When there are $n + 1$ cards, we first apply the inductive hypothesis to make the first n cards black in colour. Then we apply the inductive hypothesis to flip the last n cards. As a result, exactly 2 cards are black and the others are white. This means we can make a sequence of steps which flip exactly 2 cards.

    If $n + 1$ is even, then we are done by flipping 2 cards from white to black at a time. If $n + 1$ is odd, we flip the first k cards initially, and flip the remaining $n + 1 - k$ cards in pairs (as $n + 1 - k$ is even) so that all cards become black. So the case k being odd is done by induction.

    For even n and even k, we again proceed by induction on n. The base case $n = k$ is trivial. Suppose the case n is done. When there are $n + 2$ cards, we first apply the inductive hypothesis to make the first n cards black in colour. Then we apply the inductive hypothesis to flip the n cards except the first and the last. As a result, exactly 2 cards are black and the others are white. This means we can make a sequence of steps which flip exactly 2 cards. Since $n+2$ is even, we are done by flipping 2 cards at a time.

    Therefore, our claim holds.

33. Firstly, if the numbers lie in the ascending order $1, 2, \ldots, 100$ clockwise, then there are $\binom{100}{3} = 100 \times 33 \times 49$ clockwise triangles, which is an even number. We claim that the number of clockwise triangles must be even for any configuration.

    Suppose we swap two consecutive numbers a and b (clockwise from a to b). Only those triangles with vertices labelled by both a and b are affected. If $a < b$, then those clockwise triangles formed by a, b, c with $c < a$ or $c > b$ become anticlockwise, while those anticlockwise triangles formed by a, b, c with $a < c < b$ become clockwise. Similarly, if $a > b$, then all the 98 triangles formed by a, b, c change from clockwise to anticlockwise and vice versa. Let k be the number of clockwise triangles formed by a, b, c before the swapping. Then there are $98 - k$ such clockwise triangles after the swapping. As $98 - k$ and k have the same parity, the parity of the number of clockwise triangles is an invariant. So it must be even. In particular, it is impossible to have exactly 2017 clockwise triangles.

34. We may assume the frogs jump on the real number line. WLOG assume the initial points are $1, 2, 3, 4$. If a frog at m jumps over a frog at n, then it lands at $2n-m$, which has the same parity as m. Therefore, the points where the frogs are always consist of 2 odd numbers and 2 even numbers. However, if the distance between any two consecutive points is 2008, then the four points should have the same parity. So this situation cannot be reached.

35. We colour the cells black and white alternately. Since we add the same integer to a black square and a white square in each move, the sum of all numbers in the black squares has to be equal to the sum of all numbers in the white squares in order that it is possible to make both sums 0. We claim that this is also a sufficient condition.

    Consider any three squares with numbers a, b, c respectively such that a and b lie in two adjacent squares and b and c lie in two adjacent squares. We make the following moves:

    $$(a, b, c) \longrightarrow (a, a+b, a+c) \longrightarrow (0, b, a+c).$$

    This shows we can always make one of the numbers 0. Note that the mn squares of the chessboard can be regarded as a row of mn squares. Indeed, if we denote the squares by the coordinates (i, j) where $1 \leqslant i \leqslant m$ and $1 \leqslant j \leqslant n$, then

    $$(1,1), (1,2), \ldots, (1,n), (2,n), (2,n-1), \ldots, (2,1), (3,1), (3,2), \ldots, (3,n), \ldots$$

    is a sequence of adjacent squares. Using the above algorithm, we can make all numbers except two become 0. Since the sum of numbers in the squares of different colours is the same, the remaining two (possibly) nonzero numbers must be equal. Then we can carry out one more move to make all numbers 0.

36. The answer is 2004.

    Let the number of coins be $a_1, a_2, \ldots, a_5$ respectively. Consider any transformation. WLOG assume 4 coins from the first box are transferred to the other boxes. Then a_1 is decreased by 4 and a_j is increased by 1 for $j = 2, 3, 4, 5$. For any $2 \leqslant j \leqslant 5$, the difference $a_1 - a_j$ is decreased by 5. For any $2 \leqslant i < j \leqslant 5$, the difference $a_i - a_j$ remains unchanged. Therefore, the quantity $f_{ij} = a_i - a_j \pmod 5$ is an invariance for each pair $1 \leqslant i < j \leqslant 5$. Note that the same holds if the coins are taken from other boxes instead of the first one.

    Initially, $a_1, a_2, \ldots, a_5$ are pairwise incongruent modulo 5. From the above observation, the same holds after any number of transformation. So the minimum number of coins in four of the boxes is $0+1+2+3 = 6$, and at most $2010 - 6 = 2004$ coins can be taken away.

    The maximum value 2004 can be attained. Indeed, Martha simply needs to apply the transformation to the boxes other than the first one until this cannot be done. Since each of the remaining boxes has at most 3 coins at the end and the number of coins are pairwise incongruent modulo 5, those boxes must contain $0, 1, 2, 3$ coins respectively.

    Hence, the maximum number of coins that Martha can take away is 2004.

37. (b) Let $d = (m, n) > 1$. Suppose there is only 1 ball at the beginning. Suppose on the contrary that it is possible to make all the boxes contain an equal number of balls. Let S be the total number of balls at the end. Then S is divisible by m, the number of boxes, and hence $d \mid S$.

On the other hand, we have $S = 1 + kn$ for some $k \in \mathbb{N}$ since n balls are put in the boxes in each operation. Thus, $S \equiv 1 \pmod{d}$. This contradicts $d \mid S$ as $d > 1$.

Therefore, such a situation cannot be achieved.

38. We claim that for each pair (a_j, b_j), there must be one element not exceeding n and one element not less than $n + 1$.

Firstly, if $a_j, b_j \leqslant n$, then each of $a_1, a_2, \dots, a_j, b_j, b_{j+1}, \dots, b_n$ are less than or equal to n. There are $n+1$ such numbers, which is a contradiction. Similarly, if $a_j, b_j \geqslant n+1$, we get $n+1$ distinct numbers which are at least $n+1$ (and at most $2n$). This is impossible.

Therefore, our claim holds. It follows that

$$\begin{aligned}
&|a_1 - b_1| + |a_2 - b_2| + \cdots + |a_n - b_n| \\
&= ((n+1) + (n+2) + \cdots + (2n)) - (1 + 2 + \cdots + n) \\
&= \frac{(3n+1)n}{2} - \frac{(n+1)n}{2} = n^2.
\end{aligned}$$

39. The answer is $a = b$ or $a = b \pm 1$.

A player cannot make a move if and only both piles consist of 0 or 1 cookie. We claim that if the difference of the number of cookies in the two piles is at most 1, then the next player will lose. In other cases, the next player has a winning strategy.

We use (x, y) to denote the situation for which the first pile has x cookies and the second pile has y cookies, and let $D = |x - y|$. Consider a situation (c, d) where $D \leqslant 1$. WLOG assume the next player removes $2n$ cookies from the first pile, then the situation becomes $(c-2n, d+n)$, where $(d+n)-(c-2n) = d-c+3n \geqslant -1+3n \geqslant 2$. Therefore, after this turn, we have $D \geqslant 2$.

Consider a situation (c, d) where $D \geqslant 2$. WLOG assume $c = d + m$ where $m \geqslant 2$. Let $m = 3k - 1$, $m = 3k$ or $m = 3k + 1$ for some positive integer k. Then the next player may take $2k$ cookies from the first pile. The situation becomes $(c - 2k, d + k)$, where $D = |(c - 2k) - (d + k)| = |m - 3k|$, which is either 0 or 1.

Therefore, whenever $D = 0, 1$, after one more turn we must have $D \geqslant 2$ no matter how the next player X plays. Afterwards, the other player Y can always make D become 0 or 1. Suppose Y applies the same strategy. Since the number of cookies is strictly decreasing, it must come to one of the situations $(0, 0), (0, 1), (1, 0), (1, 1)$, each of which gives $D = 0, 1$. So the next player must be X and X will lose.

From above, Marica has a winning strategy if and only if $a = b$ or $a = b \pm 1$.

40. The answer is $a = 2^{2k}a'$ and $b = 2^{2k}b'$ for some nonnegative integer k and odd numbers a', b'.

We use (x, y) to denote the situation for which the first pile has x coins and the second pile has y coins. Let $v_2(m)$ be the largest integer k such that $2^k \mid m$. Let d be the smallest among $v_2(x)$ and $v_2(y)$. Note that the game ends when $d = 0$. Suppose we act on the even integer x. Then the situation becomes (x', y') where $x' = \frac{x}{2}$ and $y' = y + \frac{x}{2}$.

- If $v_2(x) \geqslant v_2(y) + 2$, then $v_2(x') = v_2(x) - 1$ and $v_2(y') = v_2(y)$. Then d is unchanged, and $v_2(x') \neq v_2(y')$.
- If $v_2(x) = v_2(y) + 1$, then $v_2(x') = v_2(x) - 1$ and $v_2(y') \geqslant v_2(y) + 1$. Then d is unchanged, and $v_2(x') \neq v_2(y')$.
- If $v_2(x) \leqslant v_2(y)$, then $v_2(x') = v_2(x) - 1$ and $v_2(y') = v_2(x) - 1$. Then d is decreased by 1, and $v_2(x') = v_2(y')$.

If $v_2(a) = v_2(b)$ initially, then one can only apply a move of the third type. In that case, d is decreased by 1 after each step, and we always have $v_2(x) = v_2(y)$ where (x, y) is the configuration. Thus, Alice wins if $v_2(a)$ is odd, while Bob wins if $v_2(a)$ is even.

If $v_2(a) < v_2(b)$ initially and $v_2(a)$ is odd (in particular a is even), then Alice can first apply a move (of the third type) on the pile with a coins. Then the situation becomes (x, y) where $v_2(x) = v_2(y)$ is even. As Alice becomes the second player, she can win from above.

If $v_2(a) < v_2(b)$ initially and $v_2(a)$ is even, Alice will avoid apply a move (of the third type) on the pile with a coins. After applying a move on the pile with b coins (which can always be done as b is even from $v_2(b) \geqslant 1$), the situation is basically the same. Therefore, Bob will use the same strategy as Alice. Thus, the game will never end.

Hence, Bob has a winning strategy if and only if $v_2(a) = v_2(b)$ is even.

41. Note that the numbers on the blackboard must be positive. Since

$$\frac{1}{a} + \frac{1}{b} \geqslant \frac{4}{a+b}$$

by the AM-HM inequality, the sum of reciprocals of all the numbers on the blackboard is monotonic decreasing. Therefore, the final number x satisfies

$$\frac{1}{x} \leqslant \frac{1}{1} + \frac{1}{1} + \cdots + \frac{1}{1} = n,$$

and hence $x \geqslant \frac{1}{n}$.

42. Note that the numbers on the blackboard must be positive. We have

$$\sqrt{x} + \sqrt{y} \leqslant \sqrt{2x + 2y}$$

since this is equivalent to $0 \leqslant (\sqrt{x} - \sqrt{y})^2$ by squaring both sides. So the sum of the square roots of all the numbers on the blackboard is monotonic increasing. Therefore, the final number m satisfies

$$\sqrt{m} \geqslant \sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}.$$

The function $f(x) = \sqrt{x}$ is increasing on $(0, \infty)$. It follows that

$$f(1) + f(2) + \cdots + f(n) \geqslant \int_0^n f(t)\, dt = \left[\frac{2}{3} t^{\frac{3}{2}}\right]_{t=0}^{t=n} = \frac{2}{3} n^{\frac{3}{2}}.$$

Hence, $m \geqslant \left(\frac{2}{3} n^{\frac{3}{2}}\right)^2 = \frac{4}{9} n^3$.